

3. Let (M, d) be a metric space. A point $x \in M$ is said to be a **Boundary Point** of a set A iff $B_\epsilon(x) \cap A \neq \emptyset$ and $B_\epsilon(x) \cap A^C \neq \emptyset$ for every $\epsilon > 0$.

If F is a nonempty bounded subset of $\mathbb{R}$ with usual metric, show that $\sup F$ is a boundary point of F .

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4. Consider $M = [0, 2)$ and define $d_1(x, y) = \min\{|x - y|, 0.1\}$.

- Determine all open balls in (M, d_1) .
- Are $\{0\}$ and $\{1\}$ an open set or a closed set in (M, d_1) ? Justify.
- Is $(0, 0.1]$ an open set in (M, d_1) ? Justify.

• $B_r(x) = \{y : d(y, x) < r\}$
for $r > 0.1$.
 $d(y, x) = \min\{|y - x|, 0.1\}$
 $\leq 0.1 < r$.

$\Rightarrow$ for $r > 0.1$ and $x \in M$.

$B_r(x)$ is the set M .

for $r \leq 0.1$

$d(y, x) < r \leq 0.1$
 $\Rightarrow d(y, x) < 0.1$
 $\Rightarrow d(y, x) = \min\{|y - x|, 0.1\}$
the standard metric

on $\mathbb{R}$. Also
consider any $x \in M$ and $r \leq 0.1$
 $\Rightarrow$ take set
 $= (\min\{x - r, 0\}, \min\{x + r, 2\})$

$B_r(x)$ is the set of pts in $(x - r, x + r)$ if $x - r > 0$ and $x + r < 2$
if it is $[0, x + r)$ if $x - r \leq 0$ and $x + r < 2$
and it is $(x - r, 2)$ if $x - r > 0$ and $x + r \geq 2$
it is $[0, 2)$ if $x - r \leq 0$ and $x + r \geq 2$.

second part.

~~is closed~~

Claim: $\{0\}$ is closed.

i will show this by showing $(0, 2)$ is open.

consider $x \in (0, 2)$

i can find $c, k \in \mathbb{R}$ s.t. $0 < c < x < k < 2 - k - x$ and $x - c < 0.1$ and $k - x < 0.1$

take $\epsilon = \min\{x - c, k - x\} < 0.1$

for $y \in (x - \epsilon, x + \epsilon)$

$d(y, x) < \epsilon < 0.1$

$\Rightarrow y \in (0, 2)$

as $\epsilon < 0.1$
 $|y-n| < 0.1$
 $\Rightarrow d(y,n) = |y-n|$

consider the ball
 $B_\epsilon(x) = \{y: d(y,n) < \epsilon\}$
 then as $\epsilon < 0.1$ the metric
 we know $|y-n| < \epsilon < 0.1$
 $\Rightarrow d(y,n)$

all elements in ball y

$y \in (n-\epsilon, n+\epsilon)$
 $0 < y < 2$

$|y-n| < \epsilon$
 $0 < y < 2 \Rightarrow y \in (0,2)$

$\Rightarrow$ the ball
 consider the ball
 $B_\epsilon(x) = \{y: d(y,n) < \epsilon\}$
 as ϵ is less than 0.1.
 $d(y,n) < 0.1 \Rightarrow d(y,n) = |y-n|$
 the ball is the points such that
 $|y-n| < \epsilon \Rightarrow y \in (n-\epsilon, n+\epsilon)$
 now we know that
 $y \in (0,2)$
 $\Rightarrow (0,2)$ is open set.

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what about $\{1\}$?

Part 3:

$(0, 0.1]$ is not open ball
 I will show that $\nexists \epsilon$
 such that $B_\epsilon(0.1) \subseteq (0, 0.1]$

for $\epsilon > 0.1$ we know
 $B_\epsilon = M$ which
 does not belong to $(0, 0.1]$
 and $m \notin (0, 0.1]$

for $\epsilon \leq 0.1$ the Ball B_ϵ
 contains $\neq$ is the
 interval $(n-\epsilon, n+\epsilon)$

now $\exists z \in \mathbb{R}$
 s.t. $n < z < n+\epsilon$.
 and if $z > 0.1 = n$
 then z is not part
 of $(0, 0.1]$ but belongs
 to the ball.
 $\Rightarrow$ for no hence
 proved

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